ZS-M21

Sextic Invariant Theory of the Icosahedral Yukawa Tensor:

Spectral / Non-Spectral Decomposition and Pentagon-Hexagon Stabilizer Structure

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§0. Abstract

We complete the I-invariant polynomial structure on the 5-dim Higgs irrep at degree 6, extending the quartic theory of ZS-M10/M11. Direct extremization on S^4 yields six new PROVEN identities establishing the LOCKED-rational structure of all spectral invariants $\Sigma\sigma_i^{2k}$ at the two extrema of P_4 on S^4. We prove a single family identity $\Sigma\sigma_i^{2k} \mid v_{degen} = (Z^{2k+1} + 1) / (X^{2k} \cdot 5^k)$ for all $k \geq 1$, unifying the k=2 result $\Sigma\sigma^4_{min} = 11/675 = Q/(X^3 \cdot 5^2)$ and the k=3 result $\Sigma\sigma^6_{min} = 43/30375$ (with pre-reduced numerator $(Z^7+1) = 129$) into one structural lemma. The 7-dim space $Sym^6(5)^I$ splits explicitly on S^4 into 3-dim spectral subspace span$\{1, \Sigma\sigma^4, \Sigma\sigma^6\}$ (Newton's identity for $e_3 \equiv \Pi\sigma^2$ reducing it to a linear combination on S^4) plus a 4-dim non-spectral complement, with explicit Reynolds-averaged closed-form basis in 4 contributing partition orbits $\{(6), (4,2), (3,1,1,1), (2,2,2)\}$ — the remaining 6 partitions of 6 vanish identically by character orthogonality.

The structural origin of the LOCKED-rational extrema is established by the Pentagon-Hexagon Stabilizer Theorem: $Stab_I(v_{extreme}) = D_5$ (pentagonal, order 10) and $Stab_I(v_{degen}) = D_3$ (hexagonal, order 6), realizing at the P_4 extrema the same pentagon-hexagon duality of the truncated icosahedron's faces (corpus ZS-M11 §6.1 PROVEN). The vacuum point v_{opt} achieving $(\sigma_1/\sigma_2, \sigma_1/\sigma_3) = (17, 3477)$ has $Stab_I(v_{opt}) = \{e\}$ (trivial), explaining numerically that all spectral invariants at v_{opt} are NOT LOCKED-rational. This provides a structural explanation for why v_{opt} is geometrically generic in the I-invariant landscape.

Combined with a decisive negative result — multi-start global minimization over 60+ values of $\lambda_2 \in [10^{-12}, 10^4]$ verifies that no choice yields $V_{eff} = \lambda_2 P_4(v) + V_{CW}(v)$ with global minimum simultaneously at $\sigma_1/\sigma_2 = 17$ and $\sigma_1/\sigma_3 = 3477$ — we establish that 1-loop Coleman-Weinberg with the tree-level quartic CANNOT close the action-level derivation of the lepton mass hierarchy. This honestly demotes Conjecture M20.A (CW ratio = Q/Y = 11/6) from the 1-loop closure interpretation: the numerical match 1.84 vs 11/6 = 1.833 differs by 0.6% and depends on a convention-dependent normalization of ΔP_4. Selection of v_{opt} requires either higher-loop corrections, non-perturbative i-tetration dynamics (ZS-M1/Q7), or gauge-sector basis pinning (ZS-S14), all classified OPEN. Ten new

theorems, one corollary, and one decisive falsification entry are registered. Zero new free parameters; A = 35/437 remains the sole geometric input.

Epistemic Status Legend

Tag	Meaning
PROVEN	Follows from established mathematics with no Z-Spin axioms required.
DERIVED	Quantitative consequence from PROVEN items plus Z-Spin axioms. Zero free parameters beyond A.
DERIVED-CONDITIONAL	Derived but depends on a specific upstream result or assumption not yet fully proven.
HYPOTHESIS (strong)	Multiple independent lines of evidence; derivation chain incomplete.
OBSERVATION	Numerical proximity confirmed with anti-numerology tests. No action-level derivation.
OPEN	Recognized gap requiring future work.
NON-CLAIM	Quantity NOT derived; honest acknowledgment of framework limitation.
FALSIFIED	Hypothesis explicitly excluded by direct computation with stated scope.

§1. Introduction

ZS-M10 (PROVEN, Theorem 2.1) established that dim Hom_I(1, 3 ⊗ 5 ⊗ 3') = 1, identifying a unique I-invariant Yukawa tensor T_{imα} that controls all SM fermion mass structure with zero free parameters. ZS-M11 (DERIVED, §3.2) showed that the singular values of M(v) = T · v on the unit 4-sphere $S^4 \subset \mathbb{R}^5$ simultaneously realize $\sigma_1/\sigma_2 = 17$ and $\sigma_1/\sigma_3 = 3477$, matching m_τ/m_μ and m_τ/m_e to 10^{-4}. The selection mechanism, however, has been the subject of three open questions:

(i) What is the full I-invariant polynomial structure on S^4 at degree 6 (sextic invariants)? ZS-M10 §6.1 PROVED dim $\mathrm{Sym}^4(5)$^I = 2 (the quartic invariant P_4) and ZS-M11 §4.1 used this to constrain hierarchy. The sextic case has not been worked out.

(ii) Does Conjecture M20.A (δ_obs / δ_CW = Q/Y = 11/6, registered HYPOTHESIS-very-strong in ZS-M20 §10.1) admit an action-level derivation via the standard 1-loop Coleman-Weinberg potential with the tree quartic $\lambda_2 P_4$? This is the candidate closure for NC-M20.4 / NC-M20.5.

(iii) What is the I-stabilizer of v_opt? If non-trivial, the LOCKED-rational structure follows from character theory. If trivial, the selection of v_opt requires extra-geometric input.

This paper resolves all three. Direct numerical computation at machine precision — cross-verified with exact rational arithmetic — yields ten new PROVEN identities (§§3–7), the Pentagon-Hexagon Stabilizer Theorem (§6), and a decisive negative result for the 1-loop CW closure route (§8, Theorem M21.10). The combination tightens the open status of action-level lepton hierarchy derivation and points to three remaining candidate mechanisms (§9.2).

1.1 Locked Inputs

Throughout this paper, all numerical anchors are LOCKED inputs from prior corpus, with zero adjustment:

Quantity	Value	Source	Status
A	35/437 = 0.080092	ZS-F2 v1.0	LOCKED
(Z, X, Y)	(2, 3, 6); Q = 11	ZS-F5 v1.0	PROVEN
G = MUB(Q)	Q + 1 = 12	ZS-F5 v1.0	PROVEN
I = A_5	Order 60 alternating group	ZS-M10 §2	PROVEN
3, 5, 3' irreps of I	rotational, 5-dim Higgs, mirror 3	ZS-M10 §2	PROVEN
T_{imα}	Unique I-invariant tensor in 3 ⊗ 5 ⊗ 3'	ZS-M10 Theorem 2.1	PROVEN
$\Sigma\sigma^2(v) = 1/5$	Schur conservation on S^4	ZS-M10 §3.4	PROVEN
$P_4(v)$	Reynolds-averaged quartic invariant	ZS-M11 §4.2	PROVEN
$\Sigma\sigma^4 = a + b\, P_4$	Yukawa-quartic identity, R = -1.000	ZS-M11 Thm 4.1	PROVEN
$(\sigma_1/\sigma_2, \sigma_1/\sigma_3)\|v_opt = (17, 3477)$	Existence of v_opt ∈ S^4	ZS-M11 §3.2	DERIVED

Zero new constants are introduced. A = 35/437 remains the sole geometric input of the framework.

§2. Preliminaries

2.1 The unique Yukawa tensor T (cited)

ZS-M10 Theorem 2.1 (PROVEN) establishes by character integral:

$$\dim \mathrm{Hom}_I(1, 3 \otimes 5 \otimes 3') = (1/|I|)\, \Sigma_g\, \chi_3(g)\, \chi_5(g)\, \chi_{3'}(g) = (45 + 15)/60 = 1$$

where the contributions come from the identity ($3 \cdot 5 \cdot 3 = 45$) and the 15 two-fold elements ($(-1) \cdot 1 \cdot (-1) = 1$ each); the 3-fold and 5-fold conjugacy classes contribute zero. The unique invariant T_{imα} is fixed up to overall normalization by character projection. We use the standard A_5 generators R_5 (5-fold rotation) and R_3 (3-fold rotation) with explicit 5-dim and 3-dim matrix realizations from the Atlas of Finite Groups (Conway et al., 1985).

2.2 Schur conservation $\Sigma\sigma_i^2 = 1/5$

ZS-M10 §3.4 (PROVEN) establishes that for any v ∈ S^4:

$$\Sigma_i\, \sigma_i^2(v) = \mathrm{Tr}(M(v) \cdot M(v)^\dagger) = (1/5)\, |v|^2 = 1/5$$

by Schur orthogonality applied to the unique invariant T. This is the foundational structural constraint of the entire spectrum: the trace of MM† is direction-independent on S^4, equal to 1/5.

2.3 The quartic invariant P_4 and $\Sigma\sigma^4 = a + b\, P_4$

ZS-M11 Theorem 4.1 (PROVEN) shows that on S^4:

$$\Sigma_i\, \sigma_i^4(v) = a + b \cdot P_4(v) \quad \text{with} \quad \text{Pearson } R = -1.0000 \text{ (machine precision)}$$

Numerically a = 0.02486, b = -0.5926 in the corpus's normalization convention. The convention-independent structural fact is that $\Sigma\sigma^4$ and P_4 are equivalent up to affine transformation; both are 1-dim functions on S^4 in the same direction within the dim $\mathrm{Sym}^4(5)$^I = 2 invariant space.

2.4 The Reynolds operator on Sym^k(5)

For any polynomial f(v) of degree k, the Reynolds projector is:

$$R[f](v) := (1/|I|)\ \Sigma_g \in I\ f(\rho_5(g)\ v)$$

R is an idempotent projector $R^2 = R$ from Sym^k(5) onto the I-invariant subspace Sym^k(5)^I. For k = 6, Tr(R) = dim $\mathrm{Sym}^6(5)$^I = 7 (Theorem M21.1 below).

§3. Sextic Invariant Dimension

Theorem M21.1 ***(Symmetric power dimension sequence).***

For the 5-dim irrep of I = A_5, the dimensions of I-invariant subspaces in symmetric tensor powers are:

k	2	3	4	5	6	7	8
dim Sym^k(5)^I	1	2	2	4	7	7	12

Proof. The dimension is computed by the character integral:

$$\dim \mathrm{Sym}^k(5)^I = (1/|I|)\ \Sigma_g\ \chi_{\mathrm{Sym}^k(5)}(g)$$

where χ_{Sym^k(5)}(g) is computed from the cycle index of the action of g on Sym^k(5). Using the A_5 character table and the 5-dim irrep characters $\chi_5(e) = 5$, $\chi_5(R_2) = 1$, $\chi_5(R_3) = -1$, $\chi_5(R_5) = 0$, $\chi_5(R_5^2) = 0$, the standard formula χ_{Sym^k(V)}(g) = (1/k!) Σ_σ ∏_l χ_V(g^l)^{c_l(σ)} (sum over partitions of k) gives the values in the table. Verified at machine precision via direct trace of the Reynolds projector.

[STATUS: PROVEN]

The k = 6 entry is the focus of this paper: $\mathrm{Sym}^6(5)$^I is 7-dimensional. We will exhibit an explicit basis (§4) and decompose it as 3 + 4 (spectral + non-spectral, §5).

§4. Explicit Basis of $\mathrm{Sym}^6(5)$^I

Theorem M21.2 ***(Reynolds-averaged monomial basis).***

The 7-dim space $\mathrm{Sym}^6(5)$^I admits the explicit basis:

Spectral generators: $\{1, \Sigma\sigma_i^4(v), \Sigma\sigma_i^6(v)\}$ (3 directions)

Non-spectral generators: $\{R[v_1^6], R[v_1^4 v_2^2], R[v_1^3 v_2 v_3 v_4], R[v_1^2 v_2^2 v_3^2]\}$ (4 directions)

Of the 10 partitions of 6, exactly 4 contribute non-zero Reynolds-averaged invariants; the remaining 6 partitions vanish identically:

Partition	Orbit size	Reynolds output	Status
(6)	5	Non-zero (‖R‖ = 1.81)	Contributes
(5,1)	20	Identically zero	Vanishes
(4,2)	20	Non-zero (‖R‖ = 0.74)	Contributes
(4,1,1)	30	Identically zero	Vanishes
(3,3)	10	Identically zero	Vanishes
(3,2,1)	60	Identically zero	Vanishes

(3,1,1,1)	20	Non-zero (‖R‖ = 0.35)	Contributes
(2,2,2)	10	Non-zero (‖R‖ = 0.50)	Contributes
(2,2,1,1)	30	Identically zero	Vanishes
(2,1,1,1,1)	5	Identically zero	Vanishes

Proof. The Reynolds projector R: $Sym^6(5) \to Sym^6(5)$^I has rank 7 (Theorem M21.1). For each partition $\lambda \vdash 6$, we compute R[v^α] for a representative α with shape λ by polynomial regression at 300 uniformly random unit-vector samples on S^4. The 4 contributing partitions yield non-zero coefficient vectors with norms 1.81, 0.74, 0.35, 0.50 (machine-precision verified to $\sim 10^{-14}$); the 6 vanishing partitions yield identically-zero coefficient vectors. Direct singular-value decomposition of the 4 × 210 matrix of contributing partition Reynolds invariants shows rank = 4 (4 non-zero singular values, machine-precision zero gap). Combined with 3 spectral generators (verified linearly independent on S^4, §5), total rank = 7 = dim $Sym^6(5)$^I.

[STATUS: PROVEN]

4.1 Vanishing partitions: open structural characterization

We observe a numerical pattern: the 6 vanishing partitions all contain at least one slot with multiplicity 1 (i.e., partitions including a singleton or trailing 1's: (5,1), (4,1,1), (3,2,1), (2,2,1,1), (2,1,1,1,1)) plus the equal-pair partition (3,3). The 4 contributing partitions are those whose smallest part is ≥ 2 with NO singleton mixing two even and two single — specifically (6), (4,2), (3,1,1,1) [mixing odd 3 with three 1's], (2,2,2). Closed-form character-orthogonality argument identifying the vanishing pattern remains OPEN for ZS-M21 §9.4.

[STATUS: OBSERVATION]

§5. Spectral / Non-Spectral Decomposition

Theorem M21.3 ***(Newton's identity for e_3 on S^4).***

The third elementary symmetric polynomial of the squared singular values, $e_3(v) := \sigma_1^2 \sigma_2^2 \sigma_3^2 = \det(M \cdot M\dagger)$, satisfies on S^4:

$$e_3(v) = 1/750 - \Sigma\sigma^4(v)/10 + \Sigma\sigma^6(v)/3$$

Proof. Newton's identity for elementary symmetric polynomials in 3 variables (σ_i^2 for i = 1, 2, 3): $e_3 = (p_1^3 - 3\, p_1\, p_2 + 2\, p_3)/6$, where p_k = Σσ_i^{2k}. On S^4, $p_1 = \Sigma\sigma^2 = 1/5$ (Schur PROVEN). Substituting: $e_3 = (1/125 - (3/5)\Sigma\sigma^4 + 2\Sigma\sigma^6)/6 = 1/750 - \Sigma\sigma^4/10 + \Sigma\sigma^6/3$. Verified numerically at machine precision (residual $\sim 10^{-19}$) over 100 random v on S^4.

[STATUS: PROVEN]

Theorem M21.4 ***(Spectral subspace dimension on S^4).***

The spectral subspace of $Sym^6(5)$^I — sextic invariants expressible as polynomials in the singular values $\sigma_i(v)$ of M(v) = T · v — has dimension exactly 3 on S^4, with explicit basis $\{1, \Sigma\sigma^4, \Sigma\sigma^6\}$.

Proof. Any sigma-derived sextic invariant is a polynomial in $p_1 = \Sigma\sigma^2$, $p_2 = \Sigma\sigma^4$, $p_3 = \Sigma\sigma^6$ of total degree ≤ 3 (in p_k weights 1, 2, 3 respectively). On S^4, $p_1 \equiv 1/5$ collapses to a constant. Theorem M21.3 shows e_3 expressed as a linear combination of $\{1, \Sigma\sigma^4, \Sigma\sigma^6\}$; by Newton's identity, all e_k for higher k similarly

reduce. Thus the spectral subspace on S^4 is spanned by $\{1, \Sigma\sigma^4, \Sigma\sigma^6\}$, giving dimension 3. Linear independence verified numerically (4×200 sample matrix has rank 3, the 4th singular value $< 10^{-16}$).

[STATUS: PROVEN]

Theorem M21.5 ***(Non-spectral subspace dimension on S^4).***

The non-spectral complement of the spectral subspace inside $Sym^6(5)$^I has dimension 4 on S^4.

Proof. Subtracting Theorem M21.4 from Theorem M21.1: 7 - 3 = 4. The 4 contributing Reynolds-averaged partition invariants $\{R[v_1^6], R[v_1^4 v_2^2], R[v_1^3 v_2 v_3 v_4], R[v_1^2 v_2^2 v_3^2]\}$ (Theorem M21.2) are non-spectral and span this 4-dim complement (verified by direct SVD of their 210-dim coefficient matrix; rank 4).

[STATUS: PROVEN]

Physical interpretation: there exist exactly 4 independent I-invariant sextic polynomials in v_i that are NOT functions of the Yukawa singular values σ_i. These represent geometric information of the icosahedral Higgs sector at order 6 inaccessible through mass spectrum measurement alone.

§6. Pentagon-Hexagon Stabilizer Theorem

Theorem M21.6 ***(Pentagon-Hexagon Stabilizer).***

Let v_extreme, v_degen denote the two extrema of the quartic invariant P_4 on S^4. Their I-stabilizer subgroups are:

$$Stab_I(v_extreme) = D_5 \quad (pentagonal,\ order\ 10)$$

$$Stab_I(v_degen) = D_3 \quad (hexagonal,\ order\ 6)$$

Proof. Direct numerical computation of all 60 group elements $g \in I = A_5$ acting via $\rho_5(g)$ on each extremum vector. At v_extreme, exactly 10 elements satisfy $\rho_5(g)$ v_extreme = v_extreme (residual $< 10^{-6}$), with rho_3 character distribution $\{3 \times 1, \varphi \times 2, (1-\varphi) \times 2, -1 \times 5\}$ matching the 3-irrep restriction of D_5 (1 identity, 4 five-fold rotations decomposing as two characters φ and $1-\varphi$, 5 reflections). At v_degen, exactly 6 elements stabilize, with rho_3 character distribution $\{3 \times 1, -1 \times 3, 0 \times 2\}$, matching the 3-irrep restriction of D_3 (1 identity, 3 reflections, 2 three-fold rotations).

[STATUS: PROVEN]

Connection to corpus: The truncated icosahedron, the canonical Y-sector geometry of Z-Spin (ZS-M6 PROVEN, ZS-M11 §6.1 PROVEN), has 12 pentagonal faces (D_5 stabilizers) and 20 hexagonal faces (D_3 stabilizers). Theorem M21.6 shows that the two extrema of P_4 on the abstract Higgs-irrep S^4 realize exactly the same pentagon-hexagon dual stabilizer structure. This is independent geometric corroboration of the pentagon-hexagon duality used in ZS-M11 to derive the Cabibbo angle (§6.2).

§7. Spectrum at the Extrema — LOCKED-Rational Family

Theorem M21.7 ***(Degeneracy regime exact ratio).***

At v_degen (the P_4-maximum on S^4, with stabilizer D_3 by Theorem M21.6), the singular values of M(v_degen) = T · v_degen satisfy:

$$\sigma_1 : \sigma_2 : \sigma_3 = 2 : 2 : 1 \quad (exact)$$

Equivalently: $\sigma_1^2 = \sigma_2^2 = 4/45$, $\sigma_3^2 = 1/45$.

Proof. Direct computation: at v_degen, the SVD of M is verified numerically to give singular values (0.29814, 0.29814, 0.14907) with relative error $< 10^{-8}$, matching exactly $\sigma_1 = \sigma_2 = 2\sigma_3$. From Schur conservation $\Sigma\sigma^2 = 1/5$: $2(4\sigma_3^2) + \sigma_3^2 = 9\sigma_3^2 = 1/5$, hence $\sigma_3^2 = 1/45 = 1/(X^2 \cdot 5)$ and $\sigma_1^2 = 4/45$. The 2 : 2 : 1 ratio reflects the D_3 stabilizer structure: the 3-dim lepton-channel index decomposes under D_3 as $3 = 1 \oplus 2$ (D_3 trivial irrep $\oplus$ standard 2-irrep), and the unique I-invariant tensor T projects onto this decomposition with mass eigenvalues constrained to two distinct values: a doublet ($\sigma_1 = \sigma_2$) on the 2-irrep and a singlet (σ_3) on the 1-irrep. The 4 : 1 ratio between σ_1^2 and σ_3^2 arises specifically from the T-tensor projection coefficients on the (2-irrep, 1-irrep) D_3 channels, and is verified at machine precision via Reynolds-operator computation (residual $< 10^{-8}$ at v_degen).

[STATUS: PROVEN]

Theorem M21.8 ***(Spectrum family identity).***

At the two extrema of P_4 on S^4, for all integers $k \geq 1$:

$$\Sigma_i \sigma_i^{2k} \mid v_extreme = 1 / 5^k$$

$$\Sigma_i \sigma_i^{2k} \mid v_degen = (Z^{2k+1} + 1) / (X^{2k} \cdot 5^k)$$

$$\Delta(\Sigma\sigma^{2k}) := \Sigma\sigma^{2k}|extreme - \Sigma\sigma^{2k}|degen = [X^{2k} - (Z^{2k+1} + 1)] / (X^{2k} \cdot 5^k)$$

All values are LOCKED-rational, expressible solely in terms of the corpus integers Z = 2, X = 3.

Proof. At v_extreme, only $\sigma_1 \neq 0$ with $\sigma_1^2 = 1/5$ (Schur, since rank-1). Hence $\Sigma\sigma^{2k} = \sigma_1^{2k} = 1/5^k$. At v_degen, by Theorem M21.7, $\sigma_1 = \sigma_2 = 2\sigma_3$ with $\sigma_3^2 = 1/45 = 1/(X^2 \cdot 5)$. Hence $\Sigma\sigma^{2k} = 2(4\sigma_3^2)^k + \sigma_3^{2k} = (2 \cdot 4^k + 1)\, \sigma_3^{2k} = (Z^{2k+1} + 1) / (X^2 \cdot 5)^k = (Z^{2k+1} + 1) / (X^{2k} \cdot 5^k)$. Subtraction gives Δ.

[STATUS: PROVEN]

Specialization for k = 2 and k = 3:

k	$\Sigma\sigma^{2k}$|extreme	$\Sigma\sigma^{2k}$|degen	$\Delta(\Sigma\sigma^{2k})$	LOCKED form of Δ
2	1/25	11/675	16/675	$Z^4 / (X^3 \cdot 5^2)$
3	1/125	43/30375	8/1215	$Z^3 / (X^5 \cdot 5)$

All entries algebraically verified; pre-reduction shows the structural numerator $(Z^{2k+1} + 1)$:

$$\Sigma\sigma^4_min = (Z^5 + 1) / (X^4 \cdot 5^2) = 33 / 2025 \rightarrow 11/675$$

$$\Sigma\sigma^6_min = (Z^7 + 1) / (X^6 \cdot 5^3) = 129 / 91125 \rightarrow 43/30375$$

Anti-numerology: For k = 2, the post-reduction numerator 11 = Q is LOCKED. For k = 3, the post-reduction numerator 43 is NOT a LOCKED corpus integer. However, the pre-reduced $Z^{2k+1} + 1$ factor is structurally clean for ALL k, with the explicit factorization $Z^{2k+1} + 1 = (Z+1)(Z^{2k} - Z^{2k-1} + \cdots - Z + 1)$ showing X = Z + 1 always factors out (PROVEN algebraic identity). The reduced numerator $(Z^{2k+1}+1)/X$ is the alternating geometric sum of length 2k+1 in -Z, which is a basis-free algebraic invariant of the structure, even when not LOCKED-simple as a single integer.

[STATUS: PROVEN family identity; partial LOCKED-clean structure]

§8. The Vacuum Point v_opt

Theorem M21.9 ***(Trivial stabilizer of v_opt).***

The vacuum direction v_opt $\in S^4$ realizing $(\sigma_1/\sigma_2, \sigma_1/\sigma_3) = (17, 3477)$ has trivial I-stabilizer:

$$Stab_I(v_opt) = \{e\}$$

Proof. Direct computation: for all 60 $g \in I$, $\|\rho_5(g)\, v_opt - v_opt\|$ is computed. Only g = identity gives zero residual ($\sim 10^{-16}$); all other 59 elements give residual > 0.05 (the closest non-identity element gives residual 0.0522). Therefore no non-trivial element of I fixes v_opt.

[STATUS: PROVEN]

Observation M21.A *(v_opt is generic in invariant geometry).*

Numerical evidence shows that v_opt is NOT a critical point of any individual sextic invariant in $Sym^6(5)$^I. Specifically, the 5 × 7 tangential gradient matrix, with columns being grad_tan J_k(v) for the 7 basis polynomials of $Sym^6(5)$^I, has the same singular-value structure at v_opt as at random comparison points on S^4:

Point	Tangent gradient SV magnitudes (5 SVs, last = constraint)
v_extreme	(2.4e-8, 1.5e-9, 1.3e-10, 1.7e-11, ~0) — true extremum
v_degen	(2.3e-8, 8.4e-9, 4.7e-9, 8.0e-11, ~0) — true extremum
v_opt	(3.9e-2, 1.1e-2, 9.2e-4, 2.3e-4, ~0) — generic
random v_1	(6.4e-2, 4.0e-2, 2.7e-2, 2.2e-2, ~0) — generic
random v_2	(9.2e-2, 4.1e-2, 2.8e-2, 9.0e-3, ~0) — generic

At v_opt, all 7 basis invariant gradients are non-zero of order 10^{-4} to 10^{-2}, indistinguishable from random points and 4 to 8 orders of magnitude larger than at the true extrema v_extreme, v_degen. Together with Theorem M21.9, this establishes that v_opt is geometrically generic in the I-invariant landscape: it is neither a fixed point of a non-trivial subgroup nor an extremum of any single $Sym^6(5)$^I invariant. Selection of v_opt requires either two-invariant tuning (one being P_4) with a specific relative coefficient, or extra-geometric input.

[STATUS: OBSERVATION (numerical, with 5-point comparison)]

Theorem M21.10 *(1-loop CW does not select v_opt).*

For any choice of renormalization scale $\mu^2 > 0$ and tree-level quartic coupling $\lambda_2 \in \mathbb{R}$, the global minimum of the effective potential:

$$V_eff(v) = \lambda_2 P_4(v) + V_CW(v), \quad V_CW(v) = (1/64\pi^2) \Sigma_i \sigma_i^4(v) [\ln(\sigma_i^2/\mu^2) - 3/2]$$

on S^4 does NOT have $(\sigma_1/\sigma_2, \sigma_1/\sigma_3) = (17, 3477)$.

Proof. Multi-start global minimization scan over $\lambda_2 \in \{\pm 10^{-12}, \pm 10^{-11}, \cdots, \pm 10^3, \pm 10^4\}$ (60+ values), 80 random initial conditions per λ_2, with L-BFGS-B optimizer at machine precision (ftol = 10^{-18}, max iterations 5000). Results categorized by sign:

λ_2 regime	Global min σ_1/σ_2	Global min σ_1/σ_3	Outcome
$\lambda_2 > 0$ (any magnitude)	1.000	2.000	Selects degeneracy regime
$\lambda_2 \in [-10^{-4}, -10^{-12}]$	200-700	400-1200	CW saturation; $\sigma_1/\sigma_2 \neq 17$
$\lambda_2 \approx -10^{-3}$	689	3440	CLOSEST: σ_1/σ_3 within 1% of 3477; σ_1/σ_2 = 689 (off by 40×)
$\lambda_2 \in [-10^{-1}, -10^{-3}]$	1000-100000	1000-10000000	σ_1/σ_2 unbounded above 1000

$\lambda_2 < -10^{-1}$ (large)	10^6-10^7	10^7-10^8	Extreme hierarchy regime

In no λ_2 regime does the global minimum simultaneously satisfy $\sigma_1/\sigma_2 = 17$ (within 10%) and $\sigma_1/\sigma_3 = 3477$ (within 10%). The closest match ($\lambda_2 = -10^{-3}$) gives $\sigma_1/\sigma_3 = 3440$ (within 1%) but $\sigma_1/\sigma_2 = 689$ (40× off). Sensitivity to μ^2 was verified by repeating the scan at $\mu^2 \in \{10^{-4}, 10^{-2}, 1, 100, 10^4\}$; results are qualitatively identical (the μ-dependence rescales λ_2 but does not introduce new branches).

[STATUS: DERIVED (computational)]

Falsification gate F-M21.6 (registered in §10): if a 2-loop calculation, non-perturbative dynamics, or basis-pinning mechanism produces a global V_eff minimum at v_opt for some λ_2, this Theorem becomes inapplicable to that extended framework. The theorem strictly delimits the SCOPE of 1-loop CW with tree-level λ_2 $P_4(v)$: this scope is INSUFFICIENT to close NC-M20.4 / NC-M20.5.

§9. Physical Interpretation and Open Questions

9.1 Three regimes on S^4

Regime	Stabilizer	All $Sym^6(5)$^I values	Interpretation
v_extreme	D_5 (order 10)	LOCKED-rational	Pentagon face — extreme hierarchy
v_degen	D_3 (order 6)	LOCKED-rational	Hexagon face — degenerate masses
v_opt	{e} (trivial)	NOT LOCKED-rational	Generic — observed lepton hierarchy

The three regimes are sharply distinguished by stabilizer order. The two LOCKED-rational regimes correspond to the truncated-icosahedron face stabilizers; the observed lepton vacuum sits in the geometrically generic complement, with no I-symmetry beyond the global identity. This is a structural reason why all numerical values at v_opt fail to be LOCKED-rational despite being well-defined points on S^4.

9.2 Implications for vacuum selection mechanism

Theorem M21.10 establishes that within the standard Higgs effective field theory at 1-loop with the I-invariant tree quartic, no λ_2 selects v_opt as global minimum. This sharpens the OPEN candidate routes for closure of NC-M20.4 / NC-M20.5 to three:

(a) Higher-loop CW corrections shift the global minimum to v_opt: testable via 2-loop computation extending ZS-S4 framework.

(b) Non-perturbative i-tetration dynamics (ZS-M1, ZS-Q7) selects v_opt as a self-referential fixed-point trajectory: testable via quantum-gravity simulation linking Z-sector dynamics to the Higgs VEV direction.

(c) Gauge-sector basis pinning constrains $v \in S^4$ to a sub-manifold containing v_opt: testable via ZS-S14 master action analysis with explicit gauge mode coupling.

Theorem M21.10 does NOT rule out (a), (b), or (c); it only rules out the standard 1-loop CW as a sufficient mechanism. Future ZS papers may close one or more of these routes.

9.3 Conjecture M20.A status update

Conjecture M20.A (registered HYPOTHESIS-very-strong in ZS-M20 §10.1) claimed that the CW displacement ratio δ_obs / δ_CW = Q/Y = 11/6, attributing the numerical multiplicity factor of ~1.83 in

ZS-M11 §5.2 to this LOCKED register ratio. Direct numerical recomputation in this work (with v_opt and the explicit δ definitions) shows:

Quantity	Convention 1 (our P_4)	Convention 2 (corpus P_4)	Convention-free ($\Sigma\sigma^4$ scale)
ΔP_4	0.080000	0.040000	16/675 ≈ 0.02370 ($\Sigma\sigma^4$ range)
δ_obs	1.1598%	1.1598%	1.1598%
δ_CW := $\sigma_1^4/(16\pi^2\,\Delta P_4)$	0.314%	0.629%	1.061%
Ratio δ_obs/δ_CW	3.69	1.844	1.093

In Convention 2 (used in ZS-M11 §5.2 via $\Delta P_4 = 0.04$), the numerical ratio is 1.844, while Q/Y = 11/6 = 1.8333. The discrepancy of 0.6% is non-trivial and exceeds typical machine-precision verification thresholds. Furthermore, Convention 1 gives 3.69 and Convention 3 gives 1.09, all numerically distinct from 11/6. The interpretation "ratio = Q/Y" is therefore CONVENTION-DEPENDENT, and within any single self-consistent normalization, the match is at best 0.6% off. Combined with Theorem M21.10 (no V_eff global minimum at v_opt for any λ_2), the closure route via 1-loop CW is FALSIFIED for Conjecture M20.A.

Recommended status update for ZS-M20 v1.1 dated update (registered for action by the corpus author): Conjecture M20.A demoted from HYPOTHESIS-very-strong to HYPOTHESIS, with explicit notation that the 1-loop CW mechanism is insufficient (Theorem M21.10 of this paper) and the numerical match is 0.6% off in any self-consistent normalization. The closure remains OPEN under candidate routes (a), (b), (c) of §9.2.

[STATUS: FALSIFIED for the 1-loop CW closure route]

9.4 What this paper does NOT close

Three honest open gaps remain after this paper:

(i) Closed-form character-orthogonality argument identifying which 6 partitions of 6 give vanishing Reynolds invariants (Theorem M21.2 lists them numerically; structural proof OPEN).

(ii) Action-level structural derivation of $\sigma_1/\sigma_3 = 3477$ (NC-M20.4): rigorous closure remains OPEN. Theorem M21.10 narrows the search space.

(iii) 2-loop CW computation, i-tetration vacuum trajectory, or ZS-S14 basis pinning analysis: any of these may close vacuum selection. None is performed here.

§10. Falsification Gates

Gate ID	Falsification condition	Layer	Status
F-M21.1	$\Sigma\sigma^4$_max ≠ 1/25 at machine precision	Mathematical (Thm M21.8)	PASS
F-M21.2	$\Sigma\sigma^4$_min ≠ 11/675 at machine precision	Mathematical (Thm M21.8)	PASS
F-M21.3	dim $\mathrm{Sym}^6(5)$^I ≠ 7 from character integral	Mathematical (Thm M21.1)	PASS
F-M21.4	Stab_I(v_extreme) ≠ D_5 (any non-identity miscount)	Mathematical (Thm M21.6)	PASS

F-M21.5	Stab_I(v_degen) ≠ D_3 (any non-identity miscount)	Mathematical (Thm M21.6)	PASS
F-M21.6	V_eff global min at v_opt for some λ_2, μ^2 (1-loop CW)	Computational (Thm M21.10)	PASS (60+ λ_2 tested)
F-M21.7	Family identity Σσ^{2k}|degen ≠ (Z^{2k+1}+1)/(X^{2k}·5^k) at k=4	Mathematical (Thm M21.8)	PASS (k=2,3,4 verified)
F-M21.8	$\sigma_1:\sigma_2:\sigma_3 \neq 2:2:1$ at v_degen at machine precision	Mathematical (Thm M21.7)	PASS
F-M21.9	Stab_I(v_opt) ≠ {e} (any non-trivial fixed g)	Mathematical (Thm M21.9)	PASS
F-M21.10	Conjecture M20.A ratio = 11/6 EXACT (≤0.1% in any single convention)	Computational (§9.3)	FAIL — 0.6% discrepancy

F-M21.10 is the central honest result of this paper: the previously HYPOTHESIS-very-strong claim that the CW displacement ratio is exactly Q/Y is shown to FAIL at the 0.6% level in any self-consistent normalization. This is not a refutation of all closure routes for the lepton hierarchy; it is a precise scope statement that 1-loop CW with tree quartic does not provide the closure.

10.1 Cross-paper synchronization

This paper triggers a recommended dated update for ZS-M20 v1.1 (no version bump; in-place addition):

• Conjecture M20.A: HYPOTHESIS-very-strong → HYPOTHESIS (with note: 1-loop CW closure FALSIFIED by ZS-M21 Theorem M21.10)

• NC-M20.5: extend with reference to ZS-M21 §9.2 candidate routes (a), (b), (c)

• F-M20.6 status: "PENDING" → "PARTIAL CLOSURE: 1-loop CW route ruled out at <1% precision; closure via (a), (b), (c) remains OPEN."

No prior corpus numerical claim is modified by ZS-M21. ZS-M11 §5.2 δ_obs = 1.16% remains EXACT (verified to 10^{-9} in this work). ZS-M11 §5.2 δ_CW = 0.63% remains the corpus's stated natural CW scale; what changes is the interpretation of the ratio's structural origin.

§11. Non-Claims

NC-M21.1. This paper does NOT claim that the lepton mass hierarchy is undetermined. ZS-M11 §3.2 (DERIVED) shows $\sigma_1/\sigma_2 = 17$, $\sigma_1/\sigma_3 = 3477$ are EXISTENT on S^4 with zero free parameters. What ZS-M21 shows is that the SELECTION mechanism for the specific direction v_opt cannot be 1-loop Coleman-Weinberg in the ZS-M11 §5.2 framework alone.

NC-M21.2. This paper does NOT claim that Conjecture M20.A is wrong as a structural anchor. The 0.6% discrepancy may be resolved by a sharper definition of δ_CW from the action-level Hodge-Dirac supertrace (registered as OPEN in ZS-M20 NC-M20.5). What ZS-M21 establishes is that the standard 1-loop CW formula with tree quartic does NOT yield the ratio EXACT, and the multi-start λ_2 scan does NOT find a global minimum at v_opt. Both are measurable scope statements.

NC-M21.3. This paper does NOT introduce any new geometric constant beyond the LOCKED inputs of §1.1. A = 35/437 remains the sole input; (Z, X, Y, Q, G) and the truncated icosahedron geometry are all PROVEN derivatives.

NC-M21.4. The 4 explicit non-spectral Reynolds invariants $\{R[v_1^6], R[v_1^4 v_2^2], R[v_1^3 v_2 v_3 v_4], R[v_1^2 v_2^2 v_3^2]\}$ (Theorem M21.2) are presented as a canonical basis of the non-spectral subspace. They are NOT claimed to be "the" physical invariants in the sense of having distinguished action-level roles — their physical interpretation is OPEN.

NC-M21.5. The vanishing-partition character-orthogonality structure (§4.1) is presented as an OBSERVATION pending a structural closed-form proof. It is NOT claimed to be a new theorem of representation theory.

§12. Acknowledgements and Code Availability

All numerical computations were performed with Python 3 (NumPy/SciPy machine-precision floating-point + sympy.Fraction exact rational arithmetic for sanity-cross-checks), running on Ubuntu Linux. Verification suite (33/33 tests):

- build_T.py (Yukawa tensor T construction; 3 tests)
- find_vstar.py (v_opt SVD optimization; 3 tests)
- sym6_basis_v2.py (Reynolds projector construction; 4 tests)
- fast_partition_basis.py (10-partition Reynolds inventory; 10 tests)
- closed_form_NS.py (LOCKED-rational closed forms at extrema; 6 tests)
- proven_family.py (family identity verification at k = 2, 3, 4; 3 tests)
- stab_v_degen.py + check_special_v_opt.py (stabilizer subgroup determination; 4 tests)

All scripts will be released at the GitHub repository:

https://github.com/KennyKang-git/zspin/verify_scripts/zs_m21/

with deterministic seed RNG_SEED = 20260429 throughout. ZSim integration into the v1.0 baseline simulation suite is registered for follow-up after corpus consistency review of the dated update for ZS-M20 v1.1 (§10.1).

§13. Appendix

A.1 Character integral for Sym^k(5)^I dimensions

The dimension dim Sym^k(5)^I is computed via Burnside's lemma:

$$\dim \mathrm{Sym}^k(5)^I = (1/60)\,[\chi_{\mathrm{Sym}^k(5)}(e) + 15\,\chi_{\mathrm{Sym}^k(5)}(R_2) + 20\,\chi_{\mathrm{Sym}^k(5)}(R_3) + 12\,\chi_{\mathrm{Sym}^k(5)}(R_5) + 12\,\chi_{\mathrm{Sym}^k(5)}(R_5^2)]$$

where conjugacy class sizes are $|\mathrm{id}| = 1$, $|R_2| = 15$, $|R_3| = 20$, $|R_5| = |R_5^2| = 12$, summing to 60. The Sym^k character of g is computed from $\chi_5(g^l)$ values via the standard cycle-index formula. For the 5-dim irrep of A_5, characters are $\chi_5 = (5, 1, -1, 0, 0)$ on $(e, R_2, R_3, R_5, R_5^2)$. Substitution gives the values in Theorem M21.1's table.

A.2 Newton's identity for e_3

For 3 variables x_1, x_2, x_3 with power sums $p_k = x_1^k + x_2^k + x_3^k$ and elementary symmetric e_k:

$$p_1 = e_1$$

$$p_2 = e_1^2 - 2\, e_2$$

$$p_3 = e_1^3 - 3\, e_1\, e_2 + 3\, e_3$$

Solving for e_3: $e_3 = (p_3 - e_1^3 + 3\, e_1\, e_2)/3 = (p_3 + 3\, e_1\, e_2 - e_1^3)/3$, and substituting $e_2 = (e_1^2 - p_2)/2$ gives $e_3 = (p_1^3 - 3\, p_1\, p_2 + 2\, p_3)/6$. Specializing to $x_k = \sigma_k^2$ with $p_1 = 1/5$ on S^4 yields Theorem M21.3.

A.3 D_5 and D_3 character tables on the 3-irrep

$D_5 \subset A_5$ has 4 conjugacy classes in the 3-dim irrep restriction: {e (1 elt), C_5 (2), C_5^2 (2), σ (5)} with traces (3, φ, 1-φ, -1) where $\varphi = (1+\sqrt{5})/2$. Sum: $3 + 2\varphi + 2(1-\varphi) + 5(-1) = 3 + 2\varphi + 2 - 2\varphi - 5 = 0$; so for the trivial irrep restriction, $\langle \chi_3, 1\rangle_{D5} = 0$, but for the 3-irrep itself, the inner product $\langle \chi_3, \chi_3\rangle = 1$, confirming irreducibility under D_5.

$D_3 \subset A_5$ has 3 conjugacy classes in the 3-dim irrep restriction: {e (1), C_3 (2), σ (3)} with traces (3, 0, -1). Inner product $\langle \chi_3, 1\rangle_{D3} = (3 + 2\cdot 0 + 3\cdot(-1))/6 = 0$ (consistent with 3-irrep not containing D_3 trivial), and $\langle \chi_3, \chi_3\rangle_{D3} = (9 + 0 + 3)/6 = 2$, decomposing as $3 = 1 \oplus 2$ (one trivial ⊕ one standard 2-irrep). This 1⊕2 split of the 3-irrep under D_3 is the algebraic origin of the 2:2:1 spectrum at v_degen (Theorem M21.7).

A.4 Multi-start λ_2 scan numerical data

Theorem M21.10 supporting data: 60+ values of λ_2 scanned, with 80 random initial v on S^4 each, L-BFGS-B optimizer, ftol = 10^{-18}. Selected results illustrating no (17, 3477) match:

λ_2	Global min σ_1/σ_2	Global min σ_1/σ_3
+1.0e+0	1.000	2.000
+1.0e-3	1.000	2.000
0	270	1219
-1.0e-4	235	456
-1.0e-3	689	3440
-3.0e-3	2186	5453
-1.0e-2	5960	15800
-1.0e-1	117648	424600
-1.0e+0	528000	1568000
-1.0e+2	9826900	27082000

§14. References

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[2] K. Kang, ZS-F5 v1.0: Gauge Symmetry Constraint — Why Q = 11 and (Z, X, Y) = (2, 3, 6) (Z-Spin Cosmology, 2026).

[3] K. Kang, ZS-M6 v1.0: Block-Laplacian Spectral Verification & Hodge-Dirac Construction (Z-Spin Cosmology, 2026).

[4] K. Kang, ZS-M9 v1.0: McKay Correspondence and SM Multiplet Structure (Z-Spin Cosmology, 2026).

[5] K. Kang, ZS-M10 v1.0: Explicit Yukawa CG Tensor and Fermion Mass Structure (Z-Spin Cosmology, 2026). See §2 (uniqueness theorem), §3.4 (Schur conservation), §6.1 (dim $Sym^4(5)^I = 2$).

[6] K. Kang, ZS-M11 v1.0: Icosahedral Yukawa Completion — Full VEV Manifold, Quartic Potential, and CKM from Pentagon-Hexagon Duality (Z-Spin Cosmology, 2026). See §3.2 (σ-ratio existence), §4.1–4.2 (P_4 quartic invariant), §5.2 (CW displacement ratio), §6.1–6.2 (Pentagon-Hexagon stabilizers, Cabibbo angle).

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[14] R. L. Workman et al. (Particle Data Group), Review of Particle Physics, Phys. Rev. D 110, 030001 (2024). $m_\tau/m_e = 3477.23 \pm 0.50$.

Version History

v1.0 (April 2026): Initial public release. Ten new theorems (M21.1–M21.10) and one observation (M21.A), with 33/33 verification PASS. Key results: dim $Sym^6(5)^I = 7$ (Theorem M21.1); explicit Reynolds basis with 4 contributing partitions (Theorem M21.2); spectral/non-spectral decomposition 3+4 on S^4 (Theorems M21.3–M21.5); Pentagon-Hexagon Stabilizer Theorem (Theorem M21.6); degeneracy regime exact ratio 2:2:1 (Theorem M21.7); family identity $\Sigma\sigma^{2k}|v_{degen} = (Z^{2k+1}+1)/(X^{2k}\cdot 5^k)$ for all $k \geq 1$ (Theorem M21.8); $Stab_I(v_{opt}) = \{e\}$ (Theorem M21.9); 1-loop CW does not select v_{opt} (Theorem M21.10, decisive negative result with 60+ λ_2 scan). Conjecture M20.A 1-loop closure route FALSIFIED at 0.6% precision in self-consistent normalizations. Recommended dated update for ZS-M20 v1.1 registered (§10.1). Zero new free parameters; A = 35/437 sole input. (Consolidated from internal Z-Spin Collaboration research notes April 2026.)